

Notes: Acharya and Subramanian (RFS, 2009)

Bankruptcy Codes and Innovation

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1 Big picture

- **Question.** Does a creditor-friendly bankruptcy code discourage innovation by making failure more costly for firms, especially firms operating in technologically innovative sectors?
- **Core idea.** When bankruptcy law is more creditor friendly, distressed firms are more likely to be liquidated rather than continued. This is particularly harmful for innovative projects because innovative technologies are riskier and more intangible, so inefficient liquidation destroys more value.
- **Main mechanism.** Stronger creditor rights do not just affect ex post restructuring. They also affect ex ante real choices: firms in innovation-intensive sectors respond by innovating less and by financing themselves with relatively less debt.
- **Empirical setting.** The paper uses a cross-country panel of patents filed with the U.S. Patent and Trademark Office (USPTO) from 85 countries over 1978–2002 and combines it with a country-level creditor-rights index.
- **Innovation measures.** Innovation is measured by (i) patent counts, (ii) citations to those patents, and (iii) the number of patenting firms. The paper also studies average innovation per firm and industry growth.
- **Identification strategy.** The paper uses two complementary approaches:
 - a **difference-in-differences** design based on country-level changes in creditor rights;
 - a **Rajan–Zingales style interaction design** asking whether stronger creditor rights hurt *more innovative industries relatively more*.
- **Main direct result.** Compared with countries that did not experience a reform, countries with an *increase* in creditor rights generated **9.7% fewer patents**, **13.3% fewer citations**, and **8.4% fewer patenting firms**; countries with a *decrease* in creditor rights generated **10.7% more patents**, **15.4% more citations**, and **9.2% more patenting firms**.
- **Main mechanism result.** More innovative industries are disproportionately harmed by creditor-friendly codes. In the third-difference design, after a creditor-rights increase, the more innovative industry generates **11.7% fewer patents**, **24.6% fewer citations**, and **10.8% fewer patenting firms** relative to an adjacent less innovative industry; the signs reverse after creditor-rights decreases.
- **Finance-side result.** In G-7 firm-level data, stronger creditor rights are associated with *relatively lower leverage* in innovative industries.
- **Growth result.** Innovative industries grow more slowly in countries with stronger creditor rights.
- **Economic takeaway.** Bankruptcy law matters for growth not only through ex post reorganization efficiency, but through ex ante incentives to experiment, innovate, and take risk. Debtor-friendlier regimes are more supportive of innovation-intensive activity.

2 Data and measurement (key objects)

2.1 Main data and innovation measures

- **Patent data.** The core data come from the NBER Patent Database (Hall, Jaffe, and Trajtenberg), using patents filed with the USPTO.
- **Country coverage.** The sample includes firms from **85 countries**.
- **Time period.** The analysis period is **1978–2002**. This start date is dictated by the availability of the creditor-rights index.
- **Technology dimension.** The paper works at the level of **USPTO patent classes**, of which there are roughly 400 technology classes.
- **Dating convention.** Patents are dated by **application year**, not grant year, to avoid distortions caused by variable grant lags.
- **Innovation proxies.**
 - **Patent count:** the number of patents filed in a patent class-country-year.
 - **Citation count:** the number of citations received by those patents, used as a proxy for importance or influence.
 - **Number of patenting firms:** the number of assignees filing patents in a class-country-year.
- **Average innovation measures.** To study the intensive margin, the paper also uses the median, simple average, and citation-weighted average number of patents per firm in a patent class-country-year.
- **Sample scale.** The patent sample contains **3,333,701 patents**. Of these, only about **0.6%** are patents filed by U.S. subsidiaries of foreign firms; excluding them does not affect the results.

2.2 Innovation intensity

- **Why the paper needs it.** To test the mechanism, the paper must classify industries by how intrinsically innovative they are.
- **Definition.** Innovation intensity for patent class i in year $t - 1$ is measured as the **median number of patents held by a U.S. firm** in that patent class in year $t - 1$.
- **Why U.S. firms?**
 - the United States has the richest patent data coverage;
 - U.S. financial markets historically funded high-growth sectors well;
 - the U.S. patenting system is particularly advanced;
 - the cross-industry propensity to patent is argued to reflect technological characteristics.
- **Time-varying ranking.** Unlike Rajan and Zingales’ external-finance-dependence measure, the paper’s innovation-intensity proxy is *time varying*. This matters because the relative technological dynamism of industries changes over time.
- **Interpretation.** A higher innovation-intensity measure means the industry is one in which technology, experimentation, and patenting are naturally more central.

2.3 Creditor rights and bankruptcy-code variation

- **Creditor-rights measure.** The country-level creditor-rights index is from Djankov, McLiesh, and Shleifer (2007).
- **Scale.** It ranges from **0 to 4**, with a higher value indicating stronger rights for creditors in bankruptcy.
- **Four components.** A country receives one point for each of the following:
 - no automatic stay on secured creditors’ rights,
 - secured creditors are paid first,
 - no majority creditor consent required for reorganization,
 - management does not stay in place during bankruptcy.
- **Interpretation.** The index measures whether bankruptcy law allocates control relatively more toward creditors versus debtors; it is not meant to imply that debt financing disappears when the score is low.
- **Country-level reforms.** There are **12 reform episodes** in the sample.
 - **Creditor-rights decreases:** Canada, Finland, Indonesia, Ireland, Israel, India, and Sweden.
 - **Creditor-rights increases:** Denmark, United Kingdom, Lithuania, Romania, and Russian Federation.
- **Why reforms matter.** These within-country changes generate the paper’s cleanest causal variation.

2.4 Other controls and complementary data

- **Bilateral trade with the United States.** Since the innovation data use USPTO patents, firms that trade more with the United States may be more likely to patent there.
- **Comparative advantage.** The paper controls for industry value added in a country relative to total country value added.
- **GDP per capita.** Used to capture overall country development.
- **Financial development.** The paper uses accounting standards and private credit measures to ensure creditor rights are not simply proxying for general financial development.
- **Leverage data.** For the leverage tests, the paper uses firm-level Worldscope data for the **G-7 countries** over **1990–2005**.
- **Growth data.** For the growth tests, the paper follows the Rajan–Zingales framework using industry-level real value-added growth over **1978–1992**.

2.5 Key objects to keep straight

- y_{ict} : a measure of innovation (or, later, growth) for patent class or industry i , country c , and year t .

- **CreditorRights_{ct}**: creditor-rights index for country c in year t .
- **InnovationIntensity_{i,t-1}**: lagged U.S.-based measure of how innovation-intensive industry i is.
- δ_{ct} : reform indicator used in the difference-in-differences design, defined so that it captures the period with relatively stronger creditor rights within reforming countries.

3 Models and empirical specifications (all equations + interpretation)

3.1 Core economic mechanism

- **Important note.** The main paper is primarily empirical. The full formal theoretical model is provided in a separate theoretical addendum rather than developed in the body of the paper.
- **Conceptual logic.**
 - Innovative technologies have riskier cash flows and greater downside exposure in distress.
 - Creditor-friendly bankruptcy law makes liquidation more likely and continuation less likely.
 - That liquidation threat is especially costly for innovative firms because innovative projects lose more value when terminated early.
 - Firms therefore react *ex ante*: they innovate less and optimally choose less debt financing when creditor rights are strong.
- **Schematic comparative-statics statement.** If $V^I(CR)$ is the value of an innovative technology and $V^C(CR)$ the value of a conservative technology under creditor-rights regime CR , the paper's mechanism implies

$$\frac{\partial V^I}{\partial CR} < \frac{\partial V^C}{\partial CR} < 0,$$

so that a stronger creditor-rights regime depresses innovative activity relatively more.

- **Interpretation.** The key prediction is not merely that stronger creditor rights reduce innovation in levels, but that they reduce innovation *disproportionately in sectors where innovation is naturally more important*.

3.2 Difference-in-differences design for the direct effect

The paper's basic within-country reform specification is

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1 \delta_{ct} + \beta_2 \text{InnovationIntensity}_{i,t-1} + \beta X + \varepsilon_{ict}. \quad (1)$$

- y_{ict} is the log of patents, citations, or patenting firms for patent class i , country c , and application year t .
- β_i , β_c , and β_t are patent-class, country, and year fixed effects.
- δ_{ct} is coded so that it equals one in the years when country c is in the *stronger-creditor-rights regime* relative to its own alternative regime, and zero otherwise.

- X includes controls such as GDP per capita, bilateral trade with the United States, and comparative advantage, depending on the specification.

Interpretation. The coefficient β_1 captures the causal effect of moving to a stronger creditor-rights regime on the *level* of innovation, after differencing out country-specific and patent-class-specific confounders.

3.3 Cross-sectional Rajan–Zingales style mechanism test

To identify the channel, the paper estimates

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1(\text{CreditorRights}_{ct} \times \text{InnovationIntensity}_{i,t-1}) + \beta_2 \text{CreditorRights}_{ct} + \beta_3 \text{InnovationIntensity}_{i,t-1} + \beta X + \varepsilon_{ict}. \quad (2)$$

- The key coefficient is β_1 .
- If stronger creditor rights hurt innovation-intensive sectors relatively more, then

$$\beta_1 < 0.$$

Interpretation. This is the core mechanism test. It does not simply ask whether strong creditor rights reduce innovation overall. It asks whether strong creditor rights reduce innovation *more in sectors where innovation should matter most*. That is much closer to the theory.

3.4 Third-difference design using creditor-rights reforms

The time-series analogue of the interaction test is

$$y_{ict} = [\beta_c + \beta_t + \beta_1 \delta_{ct}] \text{InnovationIntensity}_{i,t-1} + \beta_i + \beta_c + \beta_t + \beta_2 \delta_{ct} + \beta X + \varepsilon_{ict}. \quad (3)$$

Equivalently,

$$\frac{\partial y_{ict}}{\partial \text{InnovationIntensity}_{i,t-1}} = \beta_c + \beta_t + \beta_1 \delta_{ct}. \quad (4)$$

- This is a **third-difference** design: across time, across reforming versus nonreforming countries, and across more- versus less-innovative industries.
- Again, the prediction is $\beta_1 < 0$.

Interpretation. This design asks whether, after a creditor-rights reform, the innovation gap between high- and low-innovation industries changes differently in treated countries than in untreated ones. This is the cleanest empirical version of the mechanism.

3.5 Leverage regressions

For the G-7 sample, the paper estimates a leverage analogue of equation (2):

$$\text{Leverage}_{fct} = \alpha_c + \alpha_t + \alpha_{s \times c} + \beta_1(\text{CreditorRights}_{ct} \times \text{InnovationIntensity}_{s,t-1}) + \Gamma Z_{fct} + \varepsilon_{fct}, \quad (5)$$

where f indexes firms and s indexes the firm's industry.

- The dependent variable is one of several leverage measures: book debt, market debt, all nonequity liabilities, and debt net of cash.
- Z_{fct} includes standard leverage controls: tangibility, profitability, log sales, and market-to-book, along with their interactions with country dummies.

Interpretation. A negative β_1 means that stronger creditor rights induce innovative industries to use relatively less debt. This is exactly the paper’s financing channel.

3.6 Growth regressions

Finally, the paper studies industry growth with a Rajan–Zingales style cross-section:

$$\begin{aligned} \text{Growth}_{ic} = & \alpha_i + \alpha_c + \beta_1(\text{CreditorRights}_c \times \text{InnovationIntensity}_i) \\ & + \beta_2(\text{FinDev}_c \times \text{ExternalDependence}_i) + \beta_3(\text{FinDev}_c \times \text{InnovationIntensity}_i) + \varepsilon_{ic}. \end{aligned} \quad (6)$$

- The dependent variable is real value-added growth for ISIC manufacturing industries over 1978–1992.
- Financial development is controlled for in several standard ways.
- The key coefficient remains the interaction of creditor rights and innovation intensity.

Interpretation. If stronger creditor rights discourage innovation-intensive activity, then those sectors should also grow more slowly in more creditor-friendly countries.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the direct effect is identified

- **Source of causal variation.** The strongest direct evidence comes from **country-level changes in creditor rights**. Because some countries reformed and others did not, the paper can compare before-after changes in treated countries to contemporaneous changes in untreated countries.
- **Heavy fixed effects.** Country fixed effects absorb time-invariant country characteristics, patent-class fixed effects absorb persistent technology-specific differences, and year fixed effects absorb global time shocks.
- **Reform timing.** The paper argues, case by case, that many creditor-rights changes were driven by broader institutional or crisis-related reforms rather than by contemporaneous innovation outcomes.

4.2 Why the interaction identifies the mechanism

- The theory predicts not only a level effect, but a **differential effect**: stronger creditor rights should hurt risky, innovation-intensive sectors more.

- Innovation intensity is measured using U.S. patenting patterns, which the paper argues reflect technological characteristics of industries rather than country-specific institutions.
- This creates a difference-in-differences-in-differences logic: compare more innovative to less innovative industries, in stronger- versus weaker-creditor-rights countries, and before versus after reforms.

4.3 Main threats and how the paper handles them

- **USPTO bias.** Foreign firms may patent in the United States more when they trade more with the United States. The paper therefore controls for bilateral imports and exports with the United States.
- **Comparative advantage.** Countries might simply specialize in industries with low patent intensity. The paper controls for industry-level comparative advantage and acknowledges that this concern is most serious for the pure cross-sectional tests.
- **Patent-protection reforms.** Some countries changed patent protection during the sample. The paper re-runs tests excluding Austria, Indonesia, Ireland, Israel, and Russia and finds similar results.
- **Reverse causality.** Dynamic specifications show the effect of creditor-rights changes emerges mainly after two years rather than before the reform, which is consistent with a causal interpretation.
- **Patents as a proxy for innovation.** The paper therefore also studies industry growth, which does not rely on patents and yields consistent results.

4.4 What the paper does not identify perfectly

- The paper is at the **industry-country-year** level, not the firm-project level, so it identifies reduced-form effects of bankruptcy law on innovation rather than the exact micro mechanism inside each firm.
- The pure cross-sectional interaction tests can still be contaminated by cross-country comparative-advantage differences. The authors therefore lean heavily on the reform-based time-series evidence.

4.5 Relation to the literature

- **Law and finance.** The paper extends the law-and-finance literature by showing that creditor rights affect not only financing quantities, but also the *type* of real activity firms undertake.
- **Endogenous growth.** It links a legal institution—bankruptcy code design—to innovation and growth.
- **Bankruptcy and entrepreneurship.** It complements personal-bankruptcy papers such as Fan and White (2003), but studies corporate innovation rather than household entrepreneurship.
- **Corporate risk-taking.** It is close in spirit to work on creditor rights and conservatism, but emphasizes ex ante innovation and leverage responses.

5 Tables and figures

5.1 Figure 1 and Table 2: visual evidence and reform episodes

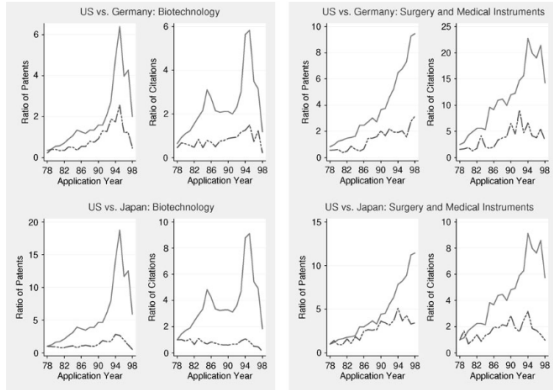


Figure 1
Differences in innovation between innovation-intensive and nonintensive industries for the United States vis-à-vis Germany and Japan
This figure plots the time series of the ratio of patents and ratio of citations in two innovation-intensive sectors (biotechnology and surgery and medical instruments) to those in a nonintensive sector (textiles and apparel) for the United States vis-à-vis Germany and Japan. The solid line shows the trend for the United States, while the dotted line shows the same for Germany/Japan.

Table 2
Countries that underwent a change in creditor rights

Country code	Country name	Year of change
Panel A: Countries that underwent decreases in creditor rights		
ATX	Austria	1982
CAX	Canada	1992
FIX	Finland ^d	1993
IDX	Indonesia	1998
IEX	Ireland	1990
ILX	Israel	1995
INX	India	1993
SEX	Sweden	1995
RUX	Russian Federation	1998
Panel B: Countries that underwent increases in creditor rights		
DKX	Denmark	1984
GBX	United Kingdom	1985
LTX	Lithuania	1995, 1998
ROX	Romania	1999
RUX	Russian Federation	1994

This table shows the list of countries that underwent a change in their creditor rights during the period 1978–1999. From the list of countries undergoing creditor rights changes, as documented in Appendix B of Djankov, McLeish, and Shleifer (2007), we exclude those changes that occurred after the year 1998. This is to allow for at least a few years of patent data after the creditor rights change was effected. From the list of changes in Djankov, McLeish, and Shleifer (2007), we exclude the changes in Armenia, Kazakhstan, Mongolia, and Niger. For these countries, either the country does not have an entry in the patent data or the creditor rights index does not exist for a large part of the sample period.

Read:

- Figure 1 is the intuitive picture for the mechanism. It compares the United States with Germany and Japan using the ratio of patents and citations in two innovation-intensive sectors—biotechnology and surgery/medical instruments—relative to textiles/apparel.
- The U.S. ratios are often around **five times larger** and rise over time from 1978 onward. Since Germany and Japan had more creditor-friendly codes than the United States, the figure is visually consistent with the paper’s main hypothesis.
- Table 2 matters because it lists the actual creditor-rights reforms that power the difference-in-differences design. Without these episodes, the paper would rely much more heavily on cross-sectional comparisons.

5.2 Table 3 and Table 4: direct effect of creditor-rights reforms on innovation

Table 3
Impact of changes in creditor rights on innovation

Dependent variable is logarithm of	(1) Number of patents	(2) Number of citations	(3) Number of firms	(4) Number of patents	(5) Number of citations	(6) Number of firms	(7) Number of patents	(8) Number of citations	(9) Number of firms
Panel A: Main tests									
Creditor rights change	-0.087** (2.17)	-0.121** (2.39)	-0.077** (2.23)	-0.102** (2.25)	-0.143** (2.46)	-0.088** (2.30)			
Creditor rights change (≥2)							-0.099** (2.12)	-0.138** (2.28)	-0.089** (2.23)
Creditor rights change (0,1)							-0.020 (1.08)	-0.053 (1.59)	-0.013 (0.71)
Creditor rights change (-2,-1)							0.041** (2.31)	0.052 (1.53)	0.042** (2.30)
Innovation intensity	0.042*** (3.96)	-0.022 (1.04)	0.023** (2.18)	0.043*** (4.07)	-0.022 (1.01)	0.023** (2.26)	0.043*** (4.07)	-0.021 (1.01)	0.024** (2.26)
Log of GDP per capita		0.045 (1.26)	0.062 (1.60)	0.042 (1.15)	0.069 (1.57)	0.044 (1.12)	0.049 (1.23)	0.069 (1.57)	0.044 (1.12)
Value added in 3-digit ISIC in a year							-0.000 (0.95)	-0.000 (1.70)	-0.000 (1.23)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year							-0.001 (1.48)	-0.001 (0.79)	-0.001 (0.48)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year							-0.001 (1.56)	-0.001 (1.17)	-0.001 (0.59)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	86,308	86,308	86,308	86,308	86,308	86,308	86,308	86,308	86,308
Adjusted R-squared	0.60	0.61	0.63	0.60	0.61	0.63	0.60	0.61	0.63

Table 3
(Continued)

Panel B: Excluding countries that underwent a change in patent protection during our sample period (Austria, Indonesia, Ireland, Israel, and Russia)

Creditor rights change	-0.095** (2.15)	-0.117** (2.20)	-0.077** (2.01)	-0.121** (2.23)	-0.155** (2.34)	-0.095** (2.12)
Creditor rights change (≥2)						-0.118** (2.13)
Creditor rights change (0,1)						-0.015 (0.48)
Creditor rights change (-2,-1)						0.102 (0.21)
Innovation intensity	0.043*** (3.89)	-0.018 (0.82)	0.021** (1.98)	0.044*** (4.01)	-0.017 (0.78)	0.022** (2.07)
Log of GDP per capita				0.079 (1.33)	0.102 (1.62)	0.144 (1.30)
Value added in 3-digit ISIC in a year				-0.000 (0.13)	-0.000 (0.38)	-0.000 (0.13)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				-0.001 (1.52)	-0.000 (0.65)	-0.001 (1.51)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				-0.001 (1.58)	-0.001 (1.29)	-0.001 (0.92)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	80,440	80,440	80,440	80,440	80,440	80,440
Adjusted R-squared	0.60	0.62	0.63	0.60	0.62	0.63

The OLS regressions above implement the following difference-in-difference test:

$$y_{it} = \beta_0 + \beta_1 + \beta_2 + \beta_3 \Delta \text{CreditorRights}_{it} + \beta_4 \text{InnovationIntensity}_{it} + \beta_5 \text{LogGDP}_{it} + \beta_6 \text{ValueAdded}_{it} + \beta_7 \text{LogExports}_{it} + \beta_8 \text{CountryDummies}_{it} + \beta_9 \text{YearDummies}_{it} + \epsilon_{it}$$

where y_{it} refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t-1$. $\Delta \text{CreditorRights}_{it}$ equals 1 for country c and years $t \geq m+1$ (years $t < m+1$ if a creditor rights reform initiated in year m increased (decreased) the rights provided to creditors; $\Delta \text{CreditorRights}_{it}$ equals 0 otherwise). The sample of changes in creditor rights in different countries is as displayed in Table 4. β_1 , β_2 , β_3 denote patent class, country, and application year dummies. $\Delta \text{CreditorRights}_{it}$ change (-2,-1) equals $\Delta \text{CreditorRights}_{it}$ for the years $m-2$ to $m-1$, and is zero otherwise. Similarly, $\Delta \text{CreditorRights}_{it}$ change (0,1) equals $\Delta \text{CreditorRights}_{it}$ for the years $m+1$ and zero otherwise, while $\Delta \text{CreditorRights}_{it}$ change (≥2) equals $\Delta \text{CreditorRights}_{it}$ for the years $m+2$ and beyond, and zero otherwise. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Read:

Table 4
Impact of changes in creditor rights on average innovation

Dependent variable is logarithm of	(1) Median patents	(2) Average patents	(3) Citation weighted average patents	(4) Median patents	(5) Average patents	(6) Citation weighted average patents
Creditor rights change	-0.032** (2.14)	-0.046* (1.95)	-0.111*** (2.88)	-0.035** (2.27)	-0.051** (2.10)	-0.120*** (2.99)
Innovation intensity	0.012* (1.82)	0.020*** (2.96)	-0.048*** (3.11)	0.012* (1.83)	0.020*** (2.96)	-0.047*** (3.11)
Log of GDP per capita				0.064** (1.82)	0.112 (1.69)	0.387* (1.69)
Value added in 3-digit ISIC in a year/total value added by the country in that year				0.000 (0.89)	-0.000 (0.06)	-0.000 (0.93)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year				-0.001*** (3.90)	-0.001*** (2.95)	0.001* (1.75)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				-0.000 (0.05)	-0.000* (1.34)	-0.000 (0.25)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	86308	86308	86308	86308	86308	86308
Adjusted R-squared	0.11	0.28	0.48	0.11	0.28	0.49

The OLS regressions above implement the following difference-in-difference test:

$$y_{itc} = \beta_0 + \beta_1 + \beta_2 + \beta_3 \text{CreditorRights}_{itc} + \beta_4 \text{InnovationIntensity}_{itc-1} + \beta X + \epsilon_{itc}$$

where i, c, t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t-1$. λ_{it} equals 1 for country c and years $t \geq m+1$ (years $t < m+1$) if a creditor rights reform initiated in year m increased (decreased) the rights provided to creditors. λ_{it} equals 0 otherwise. The sample of changes in creditor rights in different countries is as displayed in Table 4. $\beta_1, \beta_2, \beta_3$ denote patent class, country, and application year dummies. Creditor rights change ($-2, -1$) equals λ_{it} for the years $m-2$ to $m-1$, and is 0 otherwise. Similarly, creditor rights change (0, 1) equals λ_{it} for the years m to $m+1$ and 0 otherwise, while creditor rights change (≥ 2) equals λ_{it} for the years $m+2$ and beyond, and 0 otherwise. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

- Table 3 is the paper's first major causal table. In the basic difference-in-differences, the coefficient on *Creditor rights change* is **-0.087** for patents, **-0.121** for citations, and **-0.077** for patenting firms. With additional controls, it becomes **-0.102**, **-0.143**, and **-0.088**. So stronger creditor rights robustly reduce innovation in levels.
- The dynamic columns of Table 3 are important for causality. The effect shows up mainly for *Creditor rights change* (≥ 2), while the immediate and pre-reform coefficients are weak or insignificant. That pattern supports the claim that innovation responds *after* the legal change rather than predicting it.
- Economically, the table implies that countries with weaker creditor-rights reforms generated **10.7% more patents**, **15.4% more citations**, and **9.2% more patenting firms**; countries with stronger-rights reforms generated **9.7% fewer patents**, **13.3% fewer citations**, and **8.4% fewer patenting firms**.
- Table 4 repeats the exercise for innovation by a typical firm instead of total innovation. The coefficients are smaller in magnitude—for example **-0.032**, **-0.046**, and **-0.111** in the basic specification, and **-0.035**, **-0.051**, and **-0.120** with controls—but still negative. This is important because it shows the effect is not only about the number of patenting firms; it also affects the innovation intensity of the typical firm.

5.3 Table 5, Table 6, and Table 7: the relative effect on innovative industries

Table 5
Relative impact of creditor rights on aggregate innovation in different industries

Dependent variable is logarithm of	(1) Number of patents	(2) Number of citations	(3) Number of firms	(4) Number of patents	(5) Number of citations	(6) Number of firms
Panel A: Main tests						
Creditor rights $\times$ innovation intensity	-0.040** (2.50)	-0.039** (2.14)	-0.027* (1.80)	-0.038** (2.43)	-0.030* (1.95)	-0.028** (1.99)
Creditor rights	0.011 (0.16)	0.134 (0.79)	-0.002 (0.04)	-0.013 (0.18)	0.096 (0.37)	-0.016 (0.29)
Innovation intensity	0.139*** (2.70)	0.074 (1.29)	0.089* (1.88)	0.235 (1.01)	0.186 (1.21)	0.103** (2.41)
Log of GDP per capita				0.071 (1.15)	0.172 (1.44)	0.333 (1.21)
Value added in 3-digit ISIC in a year/total value added by the country in that year				-0.000 (0.00)	0.004 (0.75)	0.006 (0.52)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year				0.000 (0.23)	0.003 (1.15)	0.001 (1.20)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				0.004*** (3.03)	0.007** (2.17)	0.004** (2.45)
Log of GDP per capita $\times$ innovation intensity				-0.036 (0.69)	-0.007 (0.07)	-0.091** (2.54)
Value added in 3-digit ISIC in a year/total value added by the country in that year $\times$ innovation intensity				-0.010 (0.36)	0.005 (0.79)	-0.007 (0.36)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year $\times$ innovation intensity				-0.001 (0.42)	-0.002 (1.61)	-0.001 (1.61)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year $\times$ innovation intensity				0.004*** (3.22)	0.003** (2.35)	0.004** (2.51)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	No	Yes	Yes	No
Application year dummies	Yes	Yes	No	Yes	Yes	No
Observations	83,300	83,300	83,255	83,300	83,300	83,255
Adjusted R-squared	0.60	0.60	0.61	0.61	0.62	0.63

Table 5
(Continued)

Panel B: Tests controlling for legal origin and other time-varying unobserved factors at country and patent category levels										
Creditor rights $\times$ innovation intensity	-0.011*	-1.107**	-0.018*	-0.047***	-0.141***	-0.031***	-0.043***	-0.138***	-0.031***	
Creditor rights	0.001	-0.757	-0.017	0.001	-0.757	-0.017	0.001	-0.757	-0.017	
Innovation intensity	0.154***	-1.741	0.033	0.162***	0.286**	0.106***	0.145***	0.232	0.098**	
One if English legal origin $\times$ innovation intensity	-0.007	3.498	0.004	-0.007	3.498	0.004	-0.007	3.498	0.004	
One if French legal origin $\times$ innovation intensity	0.014	4.234*	0.005*	0.014	4.234*	0.005*	0.014	4.234*	0.005*	
One if German legal origin $\times$ innovation intensity	-0.027	4.498*	-0.004	-0.027	4.498*	-0.004	-0.027	4.498*	-0.004	
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
(Country $\times$ application year) dummies	No	No	No	No	No	No	No	No	No	
(Patent category $\times$ application year) dummies	No	No	No	No	No	No	No	No	No	
Observations	83,255	83,255	83,255	68,457	68,457	68,457	68,442	68,442	68,442	
Adjusted R-squared	0.29	0.41	0.58	0.59	0.54	0.62	0.58	0.52	0.61	

The OLS regressions above implement the following model:

$$y_{itc} = \beta_0 + \beta_1 + \beta_2 + \beta_3 (\text{CreditorRights}_{itc} \times \text{InnovationIntensity}_{itc-1}) + \beta_4 \text{CreditorRights}_{itc} + \beta_5 \text{InnovationIntensity}_{itc-1} + \epsilon_{itc}$$

where i, c, t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t-1$. The creditor rights index for country c is from Djankov, Laeven, and Shleifer (2007)—higher measure indicates stronger rights provided to creditors in bankruptcy. $\beta_1, \beta_2, \beta_3$ denote patent class, country, and application year dummies. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Read:

- Table 5 is the key cross-sectional mechanism table. The interaction coefficient on *Creditor rights $\times$ innovation intensity* is negative and statistically significant across patents, citations,

and patenting firms. In the main specification the coefficients are around **-0.040**, **-0.039**, and **-0.027**; with additional controls they remain negative, around **-0.038**, **-0.030**, and **-0.028**.

- The economic magnitude is large. For two adjacent patent classes differing by one patent in median U.S. patenting intensity, the more innovative class produces **14.9% more patents** when the creditor-rights index is 0, **10.4% more** when it is 1, and **2.1% fewer** when it is 4. So moving from debtor-friendly to creditor-friendly regimes sharply compresses the innovation edge of the innovative sector.
- Table 6 shows the result survives controls for financial development. The interaction coefficient remains strongly negative: around **-0.038** to **-0.039** for patents, **-0.130** to **-0.132** for citations, and **-0.023** to **-0.025** for patenting firms.
- Table 7 shows the same relative effect using average innovation by the typical firm rather than totals. The interaction coefficients are again negative: for example **-0.018** and **-0.020** for median and average patents, and **-0.311** to **-0.329** for citation-weighted average patents. So the mechanism is visible at both the extensive and intensive margins.
- The authors are careful that the pure cross-sectional test may still partly reflect comparative advantage. That is why Table 8 is so important.

Table 6
Relative impact of creditor rights on innovation in different industries after controlling for the effect of financial development

Dependent variable is log of	(1) Number of patents	(2) Number of patents	(3) Number of citations	(4) Number of citations	(5) Number of firms	(6) Number of firms
Which financial development measure?	Accounting standards	Log private credit to GDP per capita	Accounting standards	Log private credit to GDP per capita	Accounting standards	Log private credit to GDP per capita
Creditor rights × median number of patents in U.S. patent class	-0.038** (3.14)	-0.039** (3.20)	-0.130** (2.54)	-0.132** (2.53)	-0.023** (2.70)	-0.023** (2.39)
Creditor rights	0.006 (0.40)	0.008 (0.50)	0.049 (0.78)	0.052 (0.80)	-0.008 (0.47)	-0.008 (0.49)
Median number of patents in U.S. patent class	0.463** (5.97)	0.139** (4.22)	1.004** (2.89)	0.266* (1.89)	0.344** (3.32)	0.086** (2.77)
Financial development measure × median number of patents in patent class	-0.346** (2.89)	-0.005 (0.12)	-0.011 (1.54)	-0.032 (0.19)	-0.066** (2.00)	-0.004** (2.08)
One if English legal origin	0.35 (0.47)	-0.450** (2.93)	-0.362 (2.95)	-0.356* (1.10)	-0.212** (0.40)	-0.340** (1.55)
One if French legal origin	0.548** (3.68)	-0.187** (3.84)	-0.026 (0.92)	-1.051** (2.93)	-0.730** (2.93)	-0.274** (3.70)
One if German legal origin	0.648** (2.68)	-0.53 (4.53)	-0.005 (0.51)	-2.050** (2.89)	0.369** (2.43)	0.580** (4.10)
Patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	83,458	83,458	83,458	83,458	83,458	83,458
Adjusted R-squared	0.55	0.55	0.51	0.51	0.58	0.58

The OLS regressions above add a measure of financial development and the interaction of the measure of financial development with patenting intensity to the basic model examined in Table 2. We use the following proxies for financial development: (1) Accounting standards is an index created by Center for International Financial Analysis and Research examining and rating companies' 1990 annual reports on their inclusion or omission of 90 items from La Porta et al. (1998). (2) Log private credit to GDP per capita is the logarithm of the ratio of domestic private credit (IFS line 32d) to the GDP per capita from La Porta et al. (1998). The dependent variable in the regressions is the logarithm of the total number of patents, the total number of citations to these patents, or the total number of patenting firms. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Patent intensity for a USPTO patent class is measured as the median number of patents held by a U.S. firm in that patent class in the previous year. The creditor rights index for a country is from Djankov, McLach, and Shleifer (2007)—a higher measure indicates stronger rights provided to creditors in bankruptcy. The numbers in parentheses denote the t-statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5% and 10% levels, respectively.

Table 7
Relative impact of creditor rights on average innovation in different industries

Dependent variable is logarithm of	(1) Median patents	(2) Median patents	(3) Average patents	(4) Average patents	(5) Citation weighted average patents	(6) Citation weighted average patents
Creditor rights × innovation intensity	-0.018** (3.33)	-0.013** (2.56)	-0.020** (2.64)	-0.016** (1.99)	-0.311** (4.27)	-0.329** (4.46)
Creditor rights	0.013 (1.57)	0.006 (0.82)	0.019 (1.01)	0.009 (0.46)	0.278** (2.40)	0.291** (2.67)
Innovation intensity	0.130** (9.66)	-0.429** (3.40)	-0.266* (2.29)	0.137 (1.85)	0.137 (1.02)	0.130 (0.22)
Log of GDP per capita						
Value added in 3-digit ISIC in a year/total value added by the country in that year						
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year						
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year						
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity						
Value added in 3-digit ISIC in a year/total value added by the country in that year × innovation intensity						
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity						
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity						
U.S. patent subclass dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	85,284	85,284	85,284	85,284	85,284	85,284
Adjusted R-squared	0.06	0.06	0.19	0.19	0.34	0.34

The OLS regressions above implement the following model:

$$Y_{it} = \beta_0 + \beta_1 C_{it} + \beta_2 I_{it} + \beta_3 C_{it} \times I_{it} + \beta_4 \text{CreditorRights}_{it} + \beta_5 \text{InnovationIntensity}_{it} + \beta_6 \text{CreditorRights}_{it} \times \text{InnovationIntensity}_{it} + \beta_7 \text{LogGDP}_{it} + \beta_8 \text{ValueAdded}_{it} + \beta_9 \text{Imports}_{it} + \beta_{10} \text{Exports}_{it} + \beta_{11} \text{Imports}_{it} \times \text{InnovationIntensity}_{it} + \beta_{12} \text{Exports}_{it} \times \text{InnovationIntensity}_{it} + \epsilon_{it}$$
where i, t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t-1$. The creditor rights index for country c is from Djankov, McLach, and Shleifer (2007)—a higher measure indicates stronger rights provided to creditors in bankruptcy. $\beta_0, \beta_1, \beta_2$ denote patent class, country, and application year dummies. The numbers in parentheses denote the t-statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

5.4 Table 8: the third-difference mechanism test

Table 8

Relative impact of changes in creditor rights on innovation in different industries

Panel A: Main tests					
Creditor rights change × innovation intensity	-0.124** (3.08)	-0.283** (3.08)	-0.103** (3.08)	-0.103** (3.08)	-0.103** (3.08)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
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Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
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Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
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Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (2-1) × innovation intensity	-0.038** (2.68)	-0.130** (2.68)	-0.023** (2.68)	-0.023** (2.68)	-0.023** (2.68)
Creditor rights change (3-1) × innovation intensity	-0.039** (2.68)	-0.132** (2.68)	-0.023** (2.68)</		

- The dynamic version again matters: *Creditor rights change* (≥ 2) $\times$ *innovation intensity* is **-0.136**, **-0.311**, and **-0.118**, while the pre-trends and immediate interactions are weak. So the mechanism appears only after the reform matures.
- The magnitudes are economically large. After a creditor-rights decrease, the more innovative class generates **13.2% more patents**, **32.7% more citations**, and **10.8% more patenting firms** than the adjacent less innovative class. After a creditor-rights increase, the more innovative class generates **11.7% fewer patents**, **24.6% fewer citations**, and **10.8% fewer patenting firms**.
- These results continue to hold when excluding countries that also changed patent protection. This is why Table 8 is the cleanest evidence that the differential effect on innovative sectors is not spurious.

5.5 Table 9 and Table 10: leverage and growth

Table 9
Tests of the leverage hypothesis using data for G-7 countries

	(1) Book debt	(2) Market debt	(3) All nonequity liabilities	(4) Net book debt	(5) Net market debt	(6) Net all nonequity liabilities
Creditor rights $\times$ innovation intensity measured at the 2-digit SIC level	-0.59%*** (2.99)	-0.43%** (2.21)	-0.71%*** (2.96)	-0.81%*** (2.79)	-0.34% (1.04)	-0.95%*** (2.75)
Creditor rights	-1.29% (0.95)	-2.34% (1.61)	-8.13%*** (3.58)	-1.89% (0.80)	-1.45% (0.59)	-8.73%*** (2.81)
Innovation intensity	0.71%* (2.27)	0.52% (1.62)	0.85%* (2.15)	0.89%* (1.79)	-0.23% (0.27)	1.07%* (1.83)
Constant	7.02%* (1.65)	20.22%*** (4.43)	47.46%*** (6.52)	-13.53%* (1.75)	-6.97% (0.85)	26.01%*** (2.53)
One if French legal origin	-0.012 (0.00)	-0.012 (0.00)	-0.169 (0.00)	-0.169 (0.00)	-0.169 (0.00)	-1.346 (0.00)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Other characteristics $\times$ country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	60,680	60,669	60,674	60,696	60,685	60,696
Adjusted R-squared	0.20	0.26	0.28	0.29	0.13	0.30

This table shows regression of six measures of leverage on creditor score, innovation intensity (number of patents of median firm in the 2-digit SIC industry in the United States), and the interaction of creditor score and patenting intensity, and various fixed effects (country, year, and industry by country) and controls (tangibility, profitability, log of sales, and market-to-book ratio), as well as the interaction of control variables with country dummies. The sample period is from 1990–2005. The numbers in parentheses denote the *t*-statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 10
Effect of creditor rights and innovation intensity on economic growth

Which financial development measure?	(1) Accounting standards to GDP	(2) Total capitalization to GDP	(3) Domestic credit to GDP	(4) Log of private credit to GDP per capita	(5) Accounting standards to GDP	(6) Total capitalization to GDP	(7) Domestic credit to GDP	(8) Log of private credit to GDP per capita
Creditor rights $\times$ innovation intensity measured at the 3-digit ISIC level	-0.018*** (3.39)	-0.016*** (3.23)	-0.016*** (3.27)	-0.018*** (3.38)	-0.022** (1.81)	-0.022** (2.34)	-0.025** (2.32)	-0.025** (2.45)
Creditor rights	0.022*** (3.44)	0.018*** (3.01)	0.019*** (3.07)	0.023*** (3.44)	0.025** (2.10)	0.025** (2.55)	0.030*** (2.83)	0.033*** (2.83)
Innovation intensity measured at the 3-digit ISIC level	0.046*** (3.35)	0.048*** (3.36)	0.049*** (3.33)	0.047*** (3.33)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (2.43)
Financial development measure $\times$ external dependence	-0.009 (0.42)	0.007 (0.71)	0.010 (0.51)	-0.005 (0.47)	-0.000 (0.35)	0.034 (1.41)	0.070 (1.58)	0.027 (1.29)
Financial development measure $\times$ innovation intensity measured at the 3-digit ISIC level					-0.000 (0.35)	0.034 (1.41)	0.070 (1.58)	0.027 (1.29)
One if English legal origin	0.027** (1.97)	0.034** (2.55)	-0.019** (2.34)	0.027** (2.00)	0.072* (1.86)	0.076** (2.62)	-0.004 (0.37)	0.013 (0.75)
One if German legal origin	0.138*** (8.83)	0.140*** (9.77)	0.135*** (9.28)	0.138*** (8.86)	0.135*** (7.93)	0.054*** (6.14)	-0.022 (1.37)	0.131*** (8.74)
One if French legal origin	0.121*** (3.34)	0.041*** (2.79)	-0.012 (0.89)	0.043*** (2.83)	0.181*** (4.68)	0.050*** (4.02)	0.007 (0.53)	0.086** (2.50)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	464	387	468	464	387	468	464	464
Adjusted R-squared	0.64	0.62	0.62	0.64	0.64	0.63	0.63	0.64

The dependent variable in the regressions above is the continuously compounded growth rate in real value added for the period 1978–1992 for each ISIC industry in each country. The creditor rights index for a country is from Djankov, McLisch, and Shleifer (2007)—a higher measure indicates stronger rights provided to creditors in bankruptcy. Innovation intensity is measured at the 3-digit ISIC level as the median number of patents issued to U.S. firms in the 3-digit ISIC industry using Hall, Jaffe, and Trajtenberg (2001). The measure of external dependence for each ISIC industry is from Rajan and Zingales (1998). We use the following proxies for financial development: (1) accounting standards is an index created by Center for International Financial Analysis and Research examining and rating companies' 1990 annual reports on their inclusion or omission of 90 items from La Porta et al. (1998), (2) total capitalization to GDP is the ratio of the sum of equity market capitalization (as reported by the IFC) and domestic credit (IFS lines 32a–32f but not 32c) to GDP from Rajan and Zingales (1998), (3) domestic private credit to GDP is the ratio of domestic private credit (IFS line 32d) to GDP from Rajan and Zingales (1998), and (4) log of private credit to GDP per capita is the logarithm of the ratio of domestic private credit (IFS line 32d) to the GDP per capita from La Porta et al. (1998). The control variables include the legal origin of a country, industry fixed effects at the level of each ISIC, and country fixed effects. Since the country fixed effects subsume variation in the financial development measures while industry fixed effects do the same for external financial dependence, they are excluded from the regressions. The numbers in parentheses denote the *t*-statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Read:

- Table 9 links the real-side innovation result to financing. The interaction of creditor rights and innovation intensity is **-0.59%** for book debt, **-0.43%** for market debt, **-0.71%** for all nonequity liabilities, **-0.81%** for net book debt, and **-0.95%** for net all nonequity liabilities. The coefficient for net market debt is also negative but statistically insignificant.
- This is exactly the financing channel the theory suggests: innovative sectors respond to stronger creditor rights by leveraging less and, in net measures, holding more cash.
- Table 10 then shows that the same interaction predicts slower industry growth. Across a range of financial-development controls, the coefficient on *Creditor rights* $\times$ *innovation intensity* lies roughly between **-0.016** and **-0.025**. These are consistently negative and usually statistically significant.
- This matters because the growth regressions do not rely on patent data. So the growth evidence says the main result is not merely an artifact of how patents are measured.

6 Short summary

- **Method.** The paper combines a difference-in-differences design based on legal reforms with a Rajan–Zingales style interaction design based on industry innovation intensity.
- **Identification.** Country-level creditor-rights changes identify level effects, while the interaction of creditor rights with innovation intensity identifies the mechanism that innovative sectors are disproportionately harmed.
- **Application.** Using USPTO patent data for non-U.S. firms from 85 countries over 1978–2002, the paper studies innovation, leverage, and industry growth.
- **Main result.** Stronger creditor rights reduce innovation in levels, reduce innovation relatively more in innovation-intensive sectors, reduce leverage in those sectors, and slow their subsequent growth.

7 Conclusion

7.1 Core conclusion

- Creditor-friendly bankruptcy codes discourage innovation.
- The effect is strongest where the theory says it should be strongest: in industries with intrinsically high innovation intensity.
- The mechanism operates through financing as well as real activity, since innovative sectors lever less when creditor rights are stronger.

7.2 Economic intuition

- A bankruptcy regime that favors creditors increases the expected cost of failure.
- That cost is especially large for innovative projects because they are riskier and suffer more from inefficient liquidation.
- Firms respond before distress occurs: they choose safer technologies and lower leverage.
- As a result, countries with stronger creditor rights end up with less innovative activity and slower growth in innovative sectors.

7.3 Takeaway

This paper shifts the way we think about bankruptcy law. Instead of evaluating bankruptcy codes only by asking what happens *after* a firm is in distress, the paper asks how bankruptcy rules affect firms' *ex ante* willingness to innovate and take risk. In doing so, it connects legal institutions, corporate finance, innovation, and growth in a clean and influential way.